Chapter 2

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6.Convert the following binary numbers to decimal:
(c) 11100 (e) 10101 (g) 10111
Answer Area:
(c) $11100 = 2^2 + 2^3 + 2^4 = 28$
(e) $10101 = 2^0 + 2^2 + 2^4 = 21$
(g) $10111 = 2^0 + 2^1 + 2^2 + 2^4 = 23$

7.Convert each binary number to decimal:
(b) 101010.01 (e)1011100.10101
Answer Area:
(b) $101010.01 = 2^{-2} + 2^1 + 2^3 + 2^5 = 42.25$
(e) $1011100.10101 = 2^{-5} + 2^{-3} + 2^{-1} + 2^2 + 2^3 + 2^4 + 2^6 = 92.65625$

9.How many bits are required to represent the following decimal numbers?
(h) 205
Answer Area:
$\because 2^7 = 128 \leq 205 \leq 256 = 2^8$
$\therefore$ It needs 8 bits to represent.

13.Convert each decimal number to binary using repeated division by 2:
(h) 73
Answer Area:

	Remainder
$73/2 = 36$	1
$36/2 = 18$	0
$18/2 = 9$	0
$9/2 = 4$	1
$4/2 = 2$	0
$2/2 = 1$	0
$1/2 = 0$	1

Then the binary number is 1001001.

14.Convert each decimal fraction to binary using repeated multipliaction by 2:
(b) 0.347
Answer area:

	Carry
$0.347 \times 2 = 0.694$	0
$0.694 \times 2 = 1.388$	1
$0.388 \times 2 = 0.776$	0
$0.776 \times 2 = 1.552$	1
$0.552 \times 2 = 1.104$	1
$0.104 \times 2 = 0.208$	0
$0.208 \times 2 = 0.416$	0

Then the binary fraction is 0.010110

16.Use the direct subtraction on the following binary numbers:
(e) 1100 - 1001 (f) 11010 - 10111

Answer Area:

(e)

$$\begin{array}{rr} & 1100 \\ - & 1001 \\ \hline & 0011 \end{array}$$

Then 1100 - 1001 = 11

(f)

$$\begin{array}{rr} & 11010 \\ - & 10111 \\ \hline & 00011 \end{array}$$

Then 11010 - 10111 = 11

17.Perform the following binary multiplications:
(d) 1001×110

Answer Area:

$$\begin{array}{r} 1001 \\ \times \quad 110 \\ \hline 1001 \\ 1001 \\ \hline 110110 \end{array}$$

Then $1001 \times 110 = 110110$

18.Divide the binary numbers as indicated:
(b) 1001 ÷11
Answer Area:

$$
\begin{array}{r}
11 \\
11\overline{)1001} \\
\underline{11} \\
11 \\
\underline{11} \\
0
\end{array}
$$

Then 1001 ÷11 = 11

22.Determine the 2's complement of each binary number using either method:
(h) 00111101
Answer Area:

	00111101	Binary Number
	11000010	1's complement
+	1	Add 1
	11000011	2's complement

The 2's complement of 00111101 is 11000011.

23.Express each decimal number in binary as an 8-bit sign-magnitude number:
(c) +100 (d) -123
Answer Area:
(c)

	Remainder
$100/2 = 50$	0
$50/2 = 25$	0
$25/2 = 12$	1
$12/2 = 6$	0
$6/2 = 3$	0
$3/2 = 1$	1
$1/2 = 0$	1

Then $+100 = 01100100$

(d)

	Remainder
$123/2 = 61$	1
$61/2 = 30$	1
$30/2 = 15$	0
$15/2 = 7$	1
$7/2 = 3$	1
$3/2 = 1$	1
$1/2 = 0$	1

Then $-123 = 11111011$

24.Express each decimal number as an 8-bit number in the 1's complement form:
(c) -99 (d) +115

Answer Area:

(c)

	Remainder
$99/2 = 49$	1
$49/2 = 24$	1
$24/2 = 12$	0
$12/2 = 6$	0
$6/2 = 3$	0
$3/2 = 1$	1
$1/2 = 0$	1

The 1's complement of 1100011 is 0011100 Then $-99 = 10011100$

(d)

	Remainder
$115/2 = 57$	1
$57/2 = 28$	1
$28/2 = 14$	0
$14/2 = 7$	0
$7/2 = 3$	1
$3/2 = 1$	1
$1/2 = 0$	1

Then $+115 = 01110011$

25.Express each decimal number as an 8-bit number in the 2's complement form:
(c)+101 (d)-125

Answer Area:

(c)

	Remainder
$101/2 = 50$	1
$50/2 = 25$	0
$25/2 = 12$	1
$12/2 = 6$	0
$6/2 = 3$	0
$3/2 = 1$	1
$1/2 = 0$	1

Then $+101 = 01100101$

(d)

	Remainder
$125/2 = 62$	1
$62/2 = 31$	0
$31/2 = 15$	1
$15/2 = 7$	1
$7/2 = 3$	1
$3/2 = 1$	1
$1/2 = 0$	1

Then the 2's complement of 1111101 is 0000011.
Then $-125 = 10000011$

28.Determine the decimal value of each signed binary in the 2's complement form:
(a)10011001 (b) 01110100
Answer Area:
(a) $10011001 = 2^0 + 2^3 + 2^4 - 2^7 = -103$
(b) $01110100 = 2^2 + 2^4 + 2^5 + 2^6 = 116$

34.Perform each subtraction in the 2's complement form:
(b) 01100101 - 11101000
Answer Area:
The 2's complement of 11101000 is 00011000
Then 01100101 - 11101000 = 01100101 + 00011000 = 01111101

37.Convert each hexadecimal number to binary:
(g) $8A9D_{16}$
Answer Area:
(g) $8A9D_{16} = 1000101010011101$

38.Convert each binary number to hexadecimal:
(f)100110000010
Answer Area:
(f) $100110000010 \rightarrow 1001|1000|0010 = 982_{16}$

39.Convert each hexadecimal number to decimal:
(g) $5C2_{16}$
Answer Area:
(g) $5C2_{16} = 2 \times 16^0 + 12 \times 16^1 + 5 \times 16^2 = 1474_{10}$

40.Convert each decimal number to hexadecimal:
(g) 4019
Answer Area:
(g)

	Remainder
$4019/16 = 251$	3
$251/16 = 15$	11
$15/16 = 0$	15

Then $4019 = FB3_{16}$

42.Perform the following subtractions:
(b) $C8_{16} - 3A_{16}$
Answer Area:

$$\begin{array}{rr} & \text{C8} \\ - & \text{3A} \\ \hline & \text{8E} \end{array}$$

Then $C8_{16} - 3A_{16} = 8E_{16}$

44.Convert each decimal number to octal by repeated division by 8.
(g) 219
Answer Area:
(g)

	Remainder
$219/8 = 27$	3
$27/8 = 3$	3
$3/8 = 0$	3

Then $219 = 333$

47.Convert each of the following decimal numbers to 8421 BCD.
(k) 125
Answer Area:
(k) $125 \rightarrow 1|0010|0101$
Then $125 = 100100101_{BCD}$

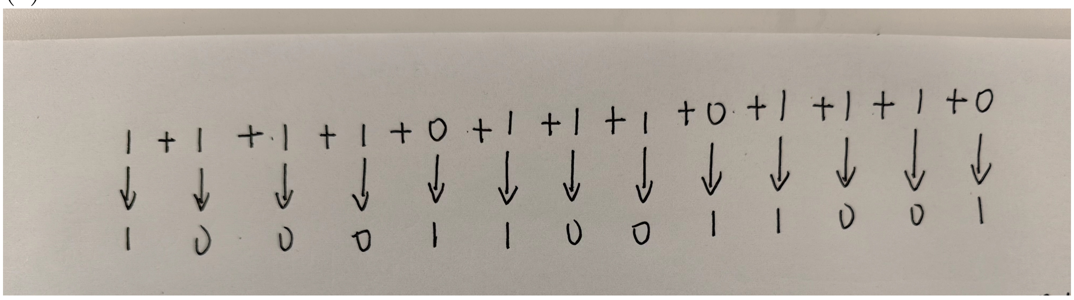
50.Convert each of the BCD numbers to decimal:
(h) 10011000
Answer Area:
(h) 10011000 $\rightarrow$ 1001|1000
Then 10011000 = 98

52.Add the following BCD numbers:
(f)

$$\begin{array}{rr} & 01100100 \\ + & 00110011 \\ \hline & 10010111 \end{array}$$

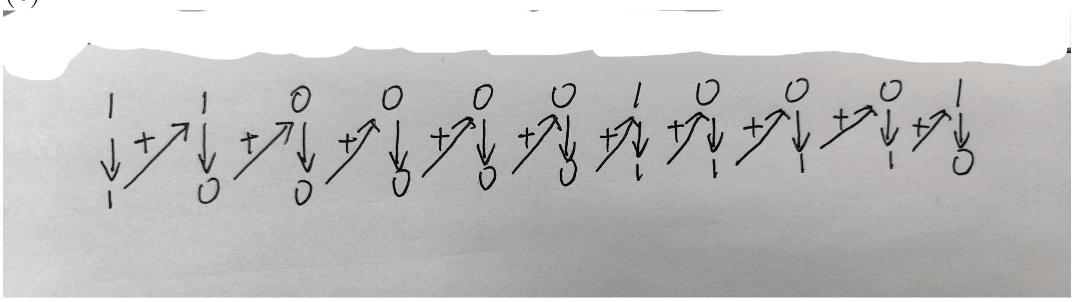
Then 01100100 + 00110011 = 10010111

56.Convert each binary number to Gray code:
(c)1111011101110
Answer Area:
(c)



Then 1111011101110 $\rightarrow$ 1000110011001

57.Convert each Gray code to binary
(c)11000010001
Answer Area:
(c)



Then 11000010001 $\rightarrow$ 10000011110

58.Determine which of the following even parity codes are in error.
(a)100110010 (b)011101010 (c)10111111010001010

Answer Area:
$\because$There are 4 1s in (a), 5 1s in (b), 10 1s in (c).
$\therefore$ (b)011101010 is wrong.